

Alfido Tech DevOps Internship

Task 3 — Infrastructure as Code with Terraform

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1. Task Overview

This task provisions basic ML infrastructure using Terraform and LocalStack, a local AWS cloud simulation tool. Three S3 buckets are created to support an ML pipeline — for datasets, model artifacts, and training logs — along with versioning enabled on the models bucket for checkpoint management.

Tech Stack

Component	Tool/Version
IaC Tool	Terraform v1.15.3
Cloud Provider (simulated)	LocalStack 3.0.0
AWS Provider	hashicorp/aws v5.100.0
Container Runtime	Docker 29.4.3
OS Environment	WSL2 Ubuntu on Windows 11
Resources	AWS S3 Buckets + Versioning

2. Project Structure

```
task3-terraform/
├── main.tf           # Provider + resource definitions
├── variables.tf      # Input variables
├── outputs.tf        # Output values
├── terraform.tfvars  # Variable values
└── README.md         # Documentation
```

3. Infrastructure Provisioned

Resource	Name	Purpose
aws_s3_bucket	alfido-ml-datasets-dev	ML training datasets
aws_s3_bucket	alfido-ml-models-dev	Model artifacts & checkpoints
aws_s3_bucket	alfido-ml-logs-dev	Training logs & metrics
aws_s3_bucket_versioning	alfido-ml-models-dev	Model version control

4. Configuration Files

4.1 variables.tf

```
variable "aws_region" {
  description = "AWS region"
```

```

    type      = string
    default   = "us-east-1"
}
variable "project_name" {
    description = "Project name prefix"
    type        = string
    default     = "alfido-ml"
}
variable "environment" {
    description = "Environment name"
    type        = string
    default     = "dev"
}
variable "localstack_endpoint" {
    description = "LocalStack endpoint URL"
    type        = string
    default     = "http://localhost:4566"
}

```

4.2 main.tf — Provider

```

provider "aws" {
    region                  = var.aws_region
    access_key              = "mock_access_key"
    secret_key              = "mock_secret_key"
    skip_credentials_validation = true
    skip_metadata_api_check  = true
    skip_requesting_account_id = true
    s3_use_path_style        = true
    endpoints {
        s3 = var.localstack_endpoint
        iam = var.localstack_endpoint
    }
}

```

4.3 main.tf — S3 Resources

```

resource "aws_s3_bucket" "ml_datasets" {
    bucket = "${var.project_name}-datasets-${var.environment}"
    tags = {
        Name      = "ML Datasets Bucket"
        Environment = var.environment
        Project    = var.project_name
    }
}
resource "aws_s3_bucket_versioning" "ml_models_versioning" {
    bucket = aws_s3_bucket.ml_models.id
    versioning_configuration {
        status = "Enabled"
    }
}

```

4.4 outputs.tf

```
output "ml_datasets_bucket" {
  value = aws_s3_bucket.ml_datasets.bucket
}
output "ml_models_bucket" {
  value = aws_s3_bucket.ml_models.bucket
}
output "ml_logs_bucket" {
  value = aws_s3_bucket.ml_logs.bucket
}
output "models_versioning_status" {
  value =
aws_s3_bucket_versioning.ml_models_versioning.versioning_configuration[0].status
}
```

5. State Handling & Variables Explanation

Terraform State

Terraform stores the current state of all provisioned resources in `terraform.tfstate`. This file is used to track what exists, plan future changes, and detect drift between real infrastructure and configuration. It should never be edited manually and should be stored securely (e.g. in an S3 backend for teams).

Variables

Variables in Terraform (`variables.tf`) allow configurations to be reusable and environment-agnostic. Values are passed via `terraform.tfvars` or CLI flags. This enables the same code to provision dev, staging, and production environments by simply changing variable values.

6. Commands Reference

Command	Description
<code>terraform init</code>	Initialize providers and backend
<code>terraform plan</code>	Preview infrastructure changes
<code>terraform apply -auto-approve</code>	Provision all resources
<code>terraform state list</code>	List all tracked resources
<code>terraform output</code>	Show output values
<code>terraform show</code>	Show full state details
<code>terraform destroy -auto-approve</code>	Destroy all resources

7. Screenshots

```
Plan: 4 to add, 0 to change, 0 to destroy.
aws_s3_bucket.ml_logs: Creating...
aws_s3_bucket.ml_datasets: Creating...
aws_s3_bucket.ml_models: Creating...
aws_s3_bucket.ml_logs: Creation complete after 1s [id=alfido-ml-logs-dev]
aws_s3_bucket.ml_models: Creation complete after 1s [id=alfido-ml-models-dev]
aws_s3_bucket.ml_datasets: Creation complete after 1s [id=alfido-ml-datasets-dev]
aws_s3_bucket_versioning.ml_models_versioning: Creating...
aws_s3_bucket_versioning.ml_models_versioning: Creation complete after 1s [id=alfido-ml-models-dev]
```

Figure 1: terraform apply — Resources being created

```
Apply complete! Resources: 4 added, 0 changed, 0 destroyed.

Outputs:

ml_datasets_bucket = "alfido-ml-datasets-dev"
ml_logs_bucket = "alfido-ml-logs-dev"
ml_models_bucket = "alfido-ml-models-dev"
models_versioning_status = "Enabled"
```

Figure 2: terraform apply — Apply complete (4 resources added)

```
hp@KVS-PC: /mnt/c/Users/hp/Documents/Alfidotech_Internship/Task3-IAC_terraform_basics$ terraform state list
aws_s3_bucket.ml_datasets
aws_s3_bucket.ml_logs
aws_s3_bucket.ml_models
aws_s3_bucket_versioning.ml_models_versioning
```

Figure 3: terraform state list — All 4 resources tracked

```
hp@KVS-PC: /mnt/c/Users/hp/Documents/Alfidotech_Internship/Task3-IAC_terraform_basics$ terraform output
ml_datasets_bucket = "alfido-ml-datasets-dev"
ml_logs_bucket = "alfido-ml-logs-dev"
ml_models_bucket = "alfido-ml-models-dev"
models_versioning_status = "Enabled"
```

Figure 4: terraform output — All bucket names confirmed

```
hp@KVS-PC: /mnt/c/Users/hp/Documents/Alfidotech_Internship/Task3-IAC_terraform_basics$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS              PORTS
NAMES
1067be14957a   localstack/localstack:3.0.0        "docker-entrypoint.sh"  42 minutes ago Up 42 minutes (healthy) 0.0.0.
0:4566->4566/tcp, [::]:4566->4566/tcp localstack
f7700b485f3a   postgres:15-alpine                "docker-entrypoint.s..." 43 hours ago   Up 3 hours          0.0.0.
0:5432->5432/tcp, [::]:5432->5432/tcp postgres-db
```

Figure 5: docker ps — LocalStack running healthy on port 4566

8. Deliverables Checklist

Deliverable	Status
main.tf — Provider + S3 resources	■ Complete
variables.tf — Input variables	■ Complete
outputs.tf — Output values	■ Complete
terraform.tfvars — Variable values	■ Complete
LocalStack running via Docker	■ Complete
terraform init — Providers initialized	■ Complete
terraform plan — 4 resources planned	■ Complete
terraform apply — 4 resources provisioned	■ Complete
terraform state list — Resources tracked	■ Complete

terraform output — Outputs verified	■ Complete
README.md with commands	■ Complete
GitHub repo pushed	■ Complete